

$e_i = (0 \dots 0 \ 1 \ 0 \ 0 \dots 0)$ : one hot vector

↳ only 1 entry is labeled as 1 all others are 0.

unit vector = vector of length 1.

$$\hookrightarrow \hat{u} = \frac{u}{\|u\|} = 1$$

$$\|u\| = \sqrt{u_1^2 + u_2^2 + \dots + u_n^2}$$

### Linear Independence

No vector can be represented as a linear combination of others.

$$\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} = \text{linearly independent}$$

$$\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\} = \text{linearly dependent.}$$

### Span

Span of a set of vectors

### Basis

A set of linearly independent vectors that span the space.

$$\text{span}(B) = \mathbb{R}^n$$

### Vector Norm

Norm is any function  $f: \mathbb{R}^n \rightarrow \mathbb{R}$  that satisfies 4 properties.

1. For all  $x \in \mathbb{R}^n$ ,  $f(x) \geq 0$

### Matrix

$$\begin{bmatrix} x_1 & x_2 & \vdots & x_n \\ y_1 & y_2 & \vdots & y_n \\ \vdots & \vdots & \ddots & \vdots \\ z_1 & z_2 & \vdots & z_n \end{bmatrix} \quad \begin{array}{l} m = \text{rows} \\ n = \text{columns} \end{array}$$

$m \times n$

$$b = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \quad b^T = [1 \ 3 \ 5]$$

$A^T = A$  = square matrix that is symmetric after transpose.

$$(A^T)^T = A$$

$$(AB)^T = B^T A^T$$

$$(A+B)^T = A^T + B^T$$

## Linear Transformation

$$y = Ax$$

$$x = \begin{pmatrix} 2 \\ 4 \end{pmatrix} \quad A = \begin{pmatrix} -1 & 3 \\ 2 & 1 \end{pmatrix} \quad y = \mathbb{R}^2$$

## Linear Equations

$$\begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ \vdots & \vdots \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \end{pmatrix}$$

$A \quad x \quad y$

$$(b_{11} \cdot x_1) + (b_{12} \cdot x_2) = y_1$$

$\vdots$

## Inverse

$$A^{-1}A = I = AA^{-1}$$

## Trace

$$\text{tr}(A) = \text{tr}(A^T)$$

$$\text{tr}(A+B) = \text{tr}(A) + \text{tr}(B)$$

$$\text{tr}(cA) = c \text{tr}(A)$$

$$\text{tr}(AB) = \text{tr}(BA)$$

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad A^{-1} = \frac{1}{|A|} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$A^{-1}$  exists if  $\det(A) \neq 0$ , if  $\det(A) = 0$ , it is singular.

## Rank

# of linearly independent vectors

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} = \text{Rank}(A) = 2$$

for any matrix  $A$ ,  $\text{column rank}(A) = \text{row rank}(A)$

## Determinant

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad |A| = ad - bc$$

$$|A| = |A^T|$$

$$|cA| = c^n |A|$$

$$|AB| = |A| |B|$$

$|A| = 0$  iff  $A$  is singular

## Diagonal Matrix

All non-diagonal values are zero.

E.g., Identity

$$AI = A = IA$$

## Eigenvalue & Eigenvector

$$Au = \lambda u, \quad u \neq 0$$

$$(\lambda I - A)u = 0, \quad u \neq 0$$

E.g.

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \quad (A - \lambda I)u = 0$$

$$\begin{pmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{pmatrix} u = 0$$

$$\begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)^2 - 1 = 0$$

$$2^2 - 2(2)(\lambda) + \lambda^2 - 1 = 0$$

$$4 - 4\lambda + \lambda^2 - 1 = 0$$

$$\lambda^2 - 4\lambda + 3 = 0$$

$$\begin{array}{cc|c} \lambda & \times & 3 \\ \lambda & \times & 1 \\ \hline \lambda^2 & & -4\lambda \end{array}$$

$$(\lambda - 3)(\lambda - 1) = 0$$

$$\lambda = 3 \text{ or } 1$$

sub  $\lambda$  to  $(\lambda I - A)u = 0$

to solve for  $u$ !